

CLINICAL INVESTIGATION

Eye

CUSTOM-MADE “NAG” EYE PLAQUES FOR ¹²⁵I BRACHYTHERAPY

SUBIR NAG, M.D.,* DIAN WANG, M.D.,* HONG WU, M.D.,† CONSTANCE J. BAUER, M.D.,*
 ROBERT B. CHAMBERS, D.O.,† AND FREDERICK H. DAVIDORF, M.D.†

*Division of Radiation Oncology and †Department of Ophthalmology, The Ohio State University Medical Center and The Arthur James Cancer Hospital, Columbus, OH

Purpose: To report our experience in treating medium-size ocular melanomas with custom-made “Nag” eye plaques.

Method: Seventy-eight patients with medium-size ocular melanomas were treated with plaques of 0.6-mm-thick, 18k gold with a radius of curvature of 12 mm to conform to the curvature of the eyeball. These plaques were custom-made in various sizes and shapes to encompass the base of the tumor and a 1-mm margin on all sides. Apertures in three wings of these plaques allowed sutures to be easily placed to hold the plaques onto the sclera. A dose of 100 Gy (before TG-43) or 85 Gy (after TG-43) was delivered with ¹²⁵I in 4 days. The dose was prescribed to the periphery of tumor at the level of tumor apex to ensure that the entire tumor volume received at least this dose.

Results: The median follow-up was 49 months (range 8–112 months). The progression-free survival (Kaplan-Meier method), overall survival, and disease-specific survival at 5 years were 93%, 86%, and 97%, respectively. Fifty of 78 (64%) patients retained useful visual acuity (20/200 or better). Thirty-one of 78 (40%) patients developed retinopathy or papillopathy. In the subgroup of 22 patients with tumor distance ≤2 mm from optic disc or macula, 12 patients (54%) developed retinopathy or papillopathy. Only 7 of these patients (32%) had useful vision (20/200 or better). Short distance to optic disc or macula was associated with significantly poor visual acuity and moderate to severe retinopathy or papillopathy (ANOVA, $p = 0.004$ and $p = 0.002$, respectively).

Conclusion: Our experience suggests that custom-made Nag plaque brachytherapy can control medium-size choroidal melanoma and that a 1-mm, rather than the standard 2-mm, minimum margin is sufficient when this plaque and prescription methods are used. Patients with tumors close (≤2 mm) to optic disc or macula have a significantly poorer visual outcome. © 2003 Elsevier Inc.

Brachytherapy, Choroidal melanoma, Iodine-125, Eye plaque, Radiotherapy.

INTRODUCTION

Enucleation has been considered the standard treatment for patients with posterior uveal melanoma. In an effort to preserve vision, episcleral plaque radiotherapy has become a commonly used alternative (1, 2). A number of radioisotopes, including radon-222, cobalt-60, gold-198, iodine-125, Ru-106/Rh-106, iridium-192, and palladium-103, have been used for episcleral radiotherapy with varying results from retrospective studies (3–19). The Collaborative Ocular Melanoma Study (COMS) Group conducted a nationwide, multi-institutional, prospective clinical trial to compare efficacy of conventional enucleation with eye plaque radiotherapy for medium-size ocular melanomas and recently reported its preliminary results (20, 21).

The COMS study used five circular plaques (of 12, 14,

16, 18, and 20-mm diameter, respectively) to deliver brachytherapy. The COMS guidelines required the use of a plaque to cover the tumor with a minimum of 2 mm on all sides. This was appropriate for standardizing therapy required in a large multi-institutional study. However, a circular plaque does not fit irregularly shaped (e.g., oval) tumors. The use of circular plaques in such instances irradiates a large area of normal retina, thereby causing a greater visual loss (Fig. 1). This is more detrimental in posteriorly located lesions because it would be difficult to avoid irradiating the macula. A number of institutions have used custom-made plaques in special patient situations (16–19). We also have used custom-made plaques, including circular, oval, kidney-shaped, and rabbit-ear plaques, to treat patients with medium-size ocular melanomas in our

Reprint requests to: Subir Nag, M.D., F.A.C.R., F.A.C.R.O., Chief of Brachytherapy, Division of Radiation Oncology, The Ohio State University Medical Center, 300 W. 10th Avenue, Columbus, OH 43210. Tel: (614) 293-3246; Fax: (614) 293-4044; E-mail: nag.1@osu.edu

Supported in part by a grant (#5P30CS16058) from the National

Cancer Institute, Bethesda, MD.

Acknowledgments—The authors wish to express their gratitude to David Carpenter for editorial assistance and to Dr. Donn Young for statistical analysis.

Received Aug 22, 2002, and in revised form Feb 28, 2003.
 Accepted for publication Mar 7, 2003.

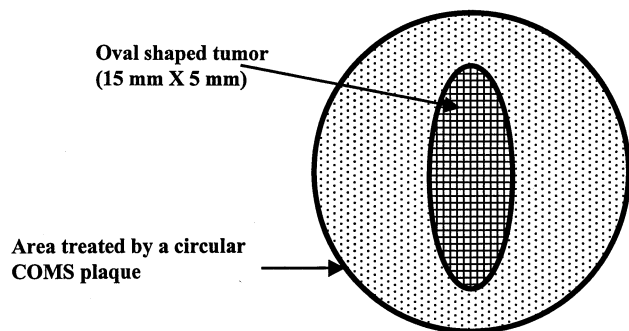


Fig. 1. The Collaborative Ocular Melanoma Study (COMS) plaque required for the irradiation of a noncircular (15×5 mm) tumor. Because a minimum of a 2-mm margin is required on all sides, and only five different COMS plaques are available, a 20-mm diameter, circular COMS plaque will be required to irradiate this tumor.

institution. To differentiate ours from others, we designated ours the Nag plaques. A detailed description is provided in the Methods section. It is to be noted that during this study, our institution participated in the COMS also; therefore, the patients treated by the Nag plaques were those who were not eligible or refused to participate in the COMS studies.

METHODS AND MATERIALS

Between 1989 and 1998, 78 patients with a clinical diagnosis of medium-size choroidal melanoma were treated

with custom-made Nag plaques at the Division of Radiation Oncology and the Department of Ophthalmology of the Ohio State University Medical Center. The clinical diagnosis of ocular melanoma was obtained from ophthalmoscopic examination, ultrasonographic evaluation, fluorescein angiography, and fundus photographs. A tumor with a basal dimension <16 mm and height <10 mm was defined as a medium-size tumor. All these patients were closely followed at approximately 6-month intervals.

The clinical variables were obtained for each patient at the time of examination. These included age, gender, medical history of arterial hypertension or diabetes mellitus, visual status (best-corrected Snellen visual acuity), and tumor data. The tumor data included the basal size of tumor, thickness and the shape of tumor, and its tumor distance to macula and optic nerve head. All patients had a metastatic workup that included liver function studies.

Radioactive Plaques

Different sizes (10–20 mm diameter) and shapes (circular, oval, kidney-shaped, and rabbit-ear) of the Nag plaques were custom-made from 0.6-mm-thick, 18k gold (Dental Ceramics, Inc., Reynoldsburg, OH) (Fig. 2). The plaques have three wings (other than the rabbit-ear plaque, which has two wings), with apertures that allow sutures to be placed to hold the plaque onto the sclera. The three-point attachment stabilizes the plaque over the globe. Although most plaques are circular or oval, we have made kidney-

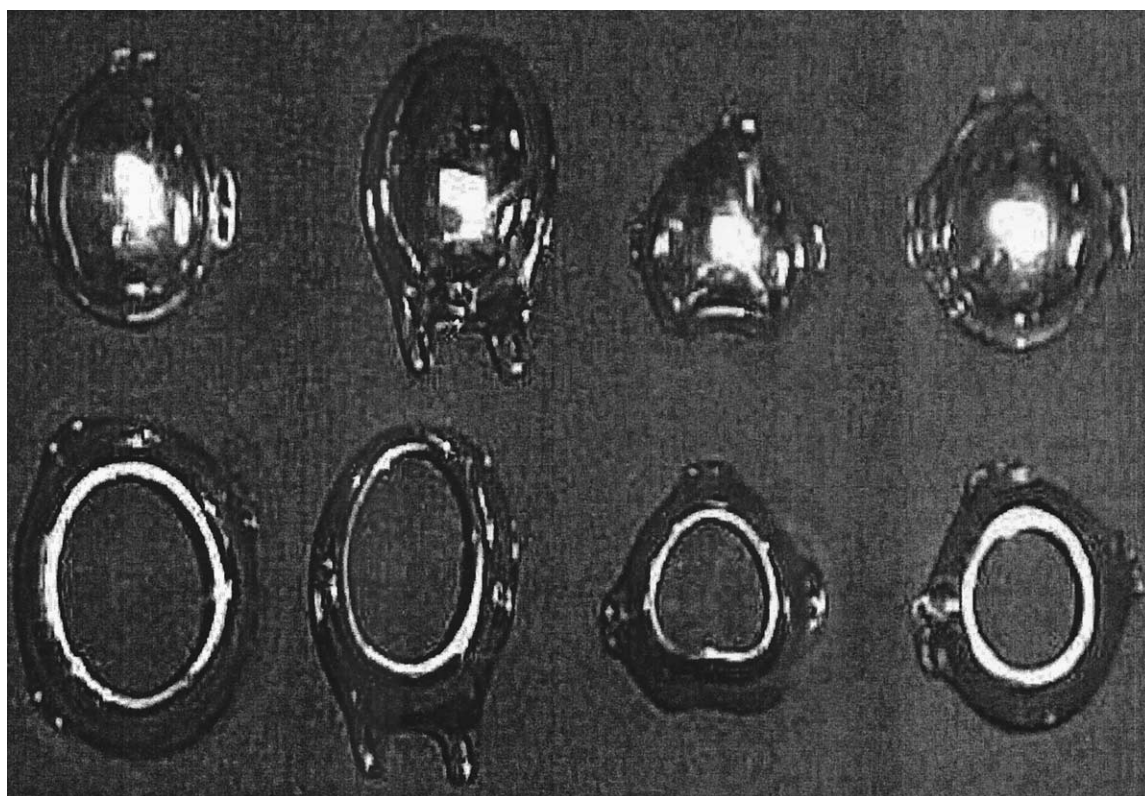


Fig. 2. Some of the different shapes and sizes of the Nag plaques. The nonradioactive dummy plaques (lower row) have a cutout in the center to allow visualization of the underlying sclera.

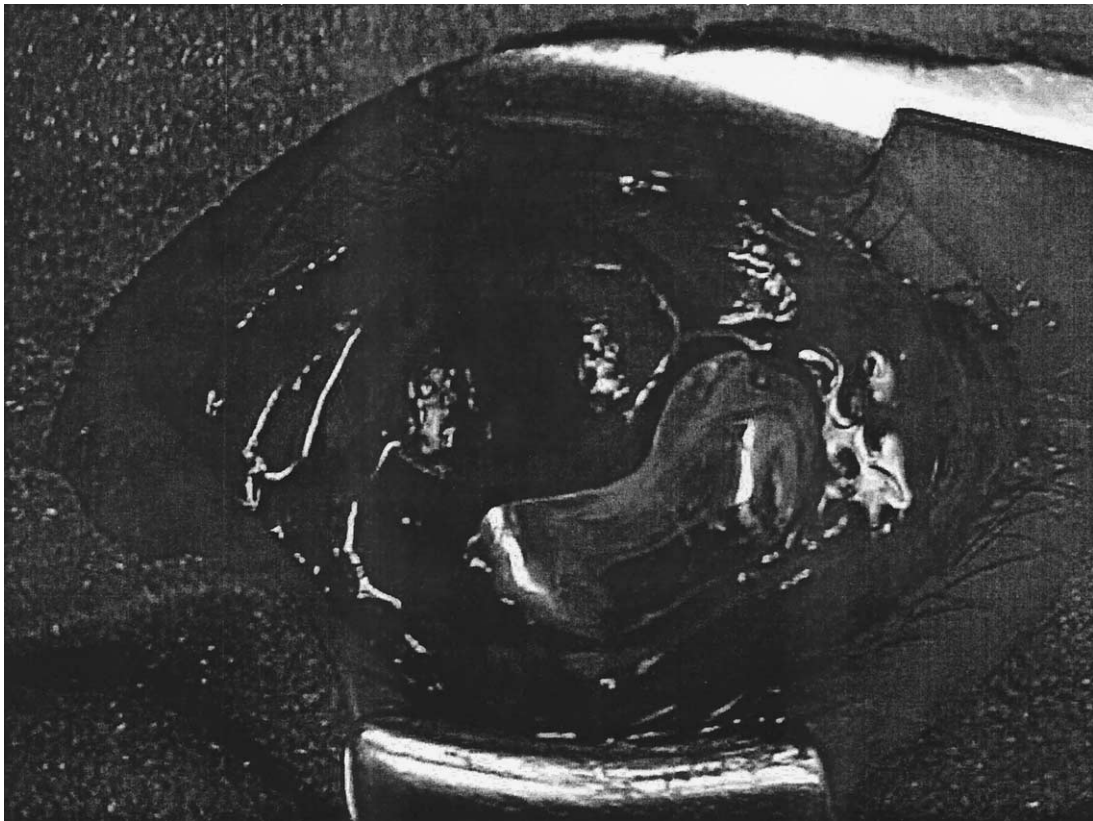


Fig. 3. A notched Nag plaque with the notch turned anteriorly to treat a tumor adjacent to the cornea.

shaped plaques for tumors close to the optic disc. By turning the concave surface anteriorly, these can also be used to treat tumors adjacent to the cornea (Fig. 3). The rabbit-ear plaques have two elongated anterior wings and are specifically designed to reach posterior-located tumors close to macula. The inner surface of the plaque is raised 2 mm from the surface of the sclera to reduce the dose to the sclera and increase the depth dose. The plaques have a small lip and a sloping surface (“unrimmed”), which make it easier for the ophthalmologist to slide the plaque in place (i.e., our plaque is user-friendly) (Fig. 4). Further, the unrimmed plaques extend the dose distribution beyond the physical edge of the plaque, hence a smaller margin can be used. Each plaque has a replica identical in size and shape but with a central cut-out to serve as a dummy (nonradioactive) plaque. The plaques were custom-made for each patient (unless a plaque of suitable shape and size already existed). We now have a large variety of plaques; therefore, we only occasionally have to fabricate a new plaque. Usually four radioactive I-125 seeds (Nycomed-Amersham, Arlington Heights, IL) of 370–740 mBq (10–20 mCi) are attached to the inner surface of the plaque using commercially available cyanoacrylate (super glue). The four seeds are arranged centrally in a quadrilateral fashion to minimize I-125 seed anisotropy. Occasionally a different seed arrangement may be used for irregularly shaped tumors to optimize the dose distribution within the tumor. The seeds can be removed after therapy by dissolving the glue with acetone. Before November 1996, a

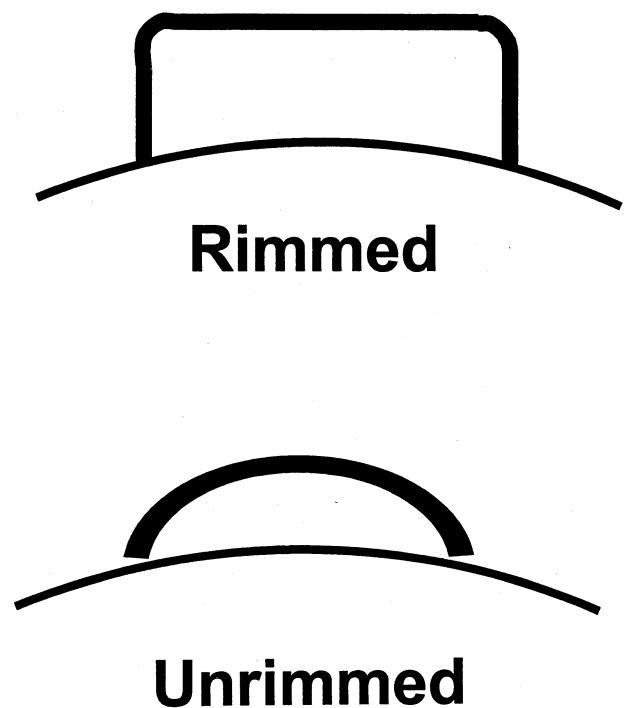


Fig. 4. The Nag plaques have a small lip and a sloping surface (“unrimmed”), which make it easier for the ophthalmologist to slide the plaque in place. Further, the unrimmed plaques extend the dose distribution beyond the physical edge of the plaque, hence a smaller margin can be used.

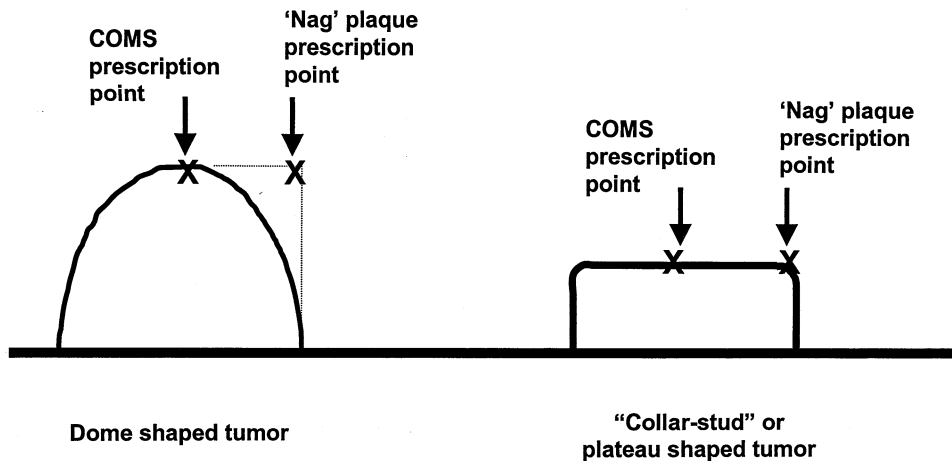


Fig. 5. Dose prescription point for Nag plaques (Nag's prescription point) is located at the same height as the apex of the tumor but laterally, perpendicular to the edge of the choroidal melanoma in its greatest basal dimension (rather than the apex of the tumor at central axis used by the Collaborative Ocular Melanoma Study). Doing so prevents any underdosage, especially for a flat or collar stud-type tumor in which the apex of the tumor is a plateau, not dome shaped.

100 Gy minimum dose was prescribed to the tumor in 4 days. After November 1996, 85 Gy was prescribed according to the American Association of Physicists in Medicine recommendation TG-43 for measuring air-kerma rates for I-125 (22). It is to be noted that the actual absorbed dose delivered to the tumor did not change as a result of the TG-43 protocol. All doses reported here have been converted to conform to the post-TG-43 protocol. We used a unique dose prescription method. Although the COMS protocol prescribed the dose to the apex at the central axis, we prescribed the dose to the periphery of tumor at the level of tumor apex to ensure that the entire tumor volume was encompassed by the prescribed dose (Fig. 5).

Plaque placement

A peritomy was performed and the area of the sclera overlying the tumor was exposed under general or local anesthesia. The overlying rectus muscle was tenotomized if necessary. The margins of the tumor were demarcated by indirect ophthalmoscopy and scleral indentation and marked (with ink) on the overlying sclera. A dummy (nonradioactive) plaque similar in size to the one to be used was then placed on the sclera over the tumor to confirm accurate localization. Scleral sutures were positioned to ensure that the plaque adequately covered the tumor with a 1-mm (rather than a 2-mm) margin. The dummy plaque was then removed and replaced with the presterilized radioactive plaque. The conjunctiva was sutured to cover the plaque. A lead rubber shield was placed over the patched globe to reduce external radiation exposure to medical caregivers and family members. With the lead patch in place, the radiation exposure was routinely <0.2 mR/h at 1 meter, thus obviating the need for any radiation isolation. When the prescribed dose was delivered to the tumor (usually in 4 days), the plaque was removed under local or general anesthesia. The rectus muscle was reinserted, if necessary, and the conjunctival incision was closed.

Follow-up

Information about the development of local tumor recurrence (measured as any degree of clinically documented growth on fundus examination, fluorescein angiography, or A-scan and B-scan ultrasonography), the development of radiation retinopathy, and papillopathy judged by ophthalmoscopy and by fluorescein angiography were recorded. Radiation retinopathy was diagnosed if any of the following features were documented clinically and angiographically: intraretinal exudation, retinal hemorrhage, dilated capillary channels, microaneurysms, perivascular sheathing, cotton-wool spots, capillary bed nonperfusion, retinal edema, cystic macular edema, or retinal neovascularization. Radiation papillopathy was diagnosed if any of the following features were documented: peripapillary exudation and hemorrhage, optic disc swelling, or optic disc neovascularization. For patients who underwent subsequent enucleation, the last visual acuity before enucleation was recorded. The complications of radiation retinopathy/papillopathy were recorded as visually notable if a decrease in visual acuity of at least three lines on the Snellen charts occurred. Evidence of metastasis was evaluated by systemic examination, liver function tests, chest radiography, computed tomography, magnetic resonance imaging or ultrasonography, and tissue biopsy if imaging studies demonstrated suspicious lesions. One or more of the following determined the survival status of each patient:

- follow-up examination at the Department of Ophthalmology or Division of Radiation Oncology;
- data obtained from the referring ophthalmologist, the family physicians, or internists; or
- data from oncologists who regularly saw the patients unable to return to our hospitals.

Occasionally, telephone interviews with the patient or close family members were also used.

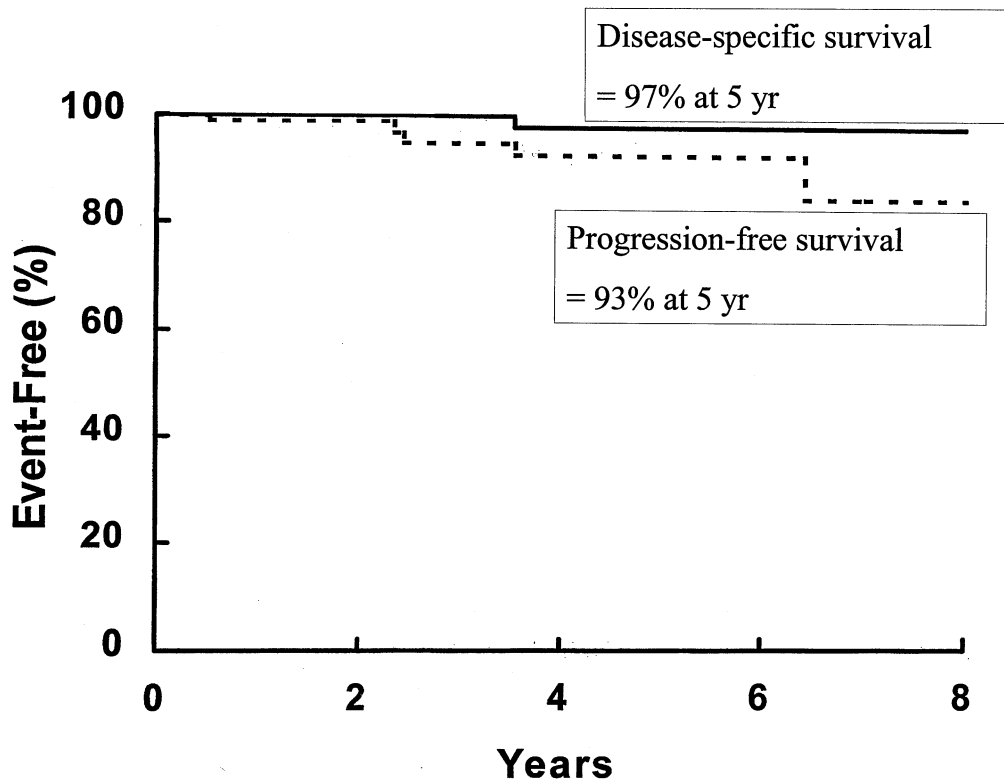


Fig. 6. The progression-free survival interval was 93% at 5 years (Kaplan-Meier method), and the disease-specific survival was 97% at 5 years. The corresponding results at 8 years were 85% and 97%, respectively.

Analysis

Four end points comprised the major outcomes of the study:

1. moderate to severe radiation retinopathy/papillopathy;
2. postradiation visual acuity;
3. local tumor recurrence; and
4. distant metastasis.

The effect of several individual predictor variables was analyzed by means of analysis of variance (ANOVA) or chi-square. The visual acuity on the Snellen chart at the last follow-up was used for analysis; 20/200 or better was clinically determined as useful vision.

Clinical results

Seventy-eight patients (46 white females and 32 white males) with a clinical diagnosis of medium-size ocular melanoma between 1989 and 1998 received custom-made Nag plaque radiotherapy. Seventy-eight eyes (42 right and 36 left) were treated. The median time of follow-up was 49 months (range 8–112 months). The mean age of the patients was 64 years (range 35–75 years). The tumor dose at the prescription point (Nag's point, Fig. 5) ranged from 64 to 94 Gy (median 85 Gy). The dose at 5 mm varied from 36 to 198 Gy (median 85 Gy), whereas the scleral dose varied from 192 to 1,068 Gy (median 458 Gy). One patient, who had a diagnosis of liver metastasis 1 month after his epis-

cleral plaque radiotherapy, received subsequent salvage chemotherapy and radiotherapy and died 36 months later. Four patients had clinical impression of local progression at follow-up (mean time 47 months; range 26–76 months) after their episcleral plaque radiotherapy. Among these 4 patients with local failure, 1 patient had a salvage laser ablation surgery for documented growth and survived without disease. Another patient had an enucleation for extrascleral extension and survived without disease. Another patient had an excisional biopsy for documented growth; however, pathology did not reveal any viable tumor cells. The fourth patient was closely followed for possible extrascleral extension. The progression-free survival (Kaplan-Meier method), overall survival, and disease-specific survival at 5 years were 93%, 86%, and 97% respectively (Fig. 6). Fifty of 78 patients (64%) had maintained their visual acuity at 20/200 or better at the last follow-up. Of the 78 patients, 31 (40%) developed moderate to severe retinopathy or papillopathy. Two patients required enucleations because they developed severe neovascular glaucoma at 6 months and 12 months, respectively.

The distance between tumor and the optic disc and macula was ≤ 2 mm in 22 of 78 (28%) patients. In the subgroup of 22 patients with ≤ 2 mm tumor distance to optic disc or macula, 12 patients (54%) developed retinopathy or papillopathy. Only 7 of these patients (32%) had useful vision (20/200 or better). Short distance to optic disc or macula

Table 1. Relationship of significant retinopathy or papillopathy and distance to macula or optic nerve disc

	No retinopathy/ papillopathy	Had retinopathy/ papillopathy	<i>p</i> value (ANOVA)
% of patients	57%	43%	
Distance to macula (mm)	7.6 ± 5.3	3.8 ± 4.0	0.002
Distance to optic disc (mm)	8.3 ± 4.8	4.9 ± 4.2	0.004

Abbreviation: ANOVA = analysis of variance.

was significantly associated with poor visual acuity and moderate to severe retinopathy or papillopathy (ANOVA, $p = 0.004$ and $p = 0.002$, respectively) (Table 1). However, there was no significant association between the radioactive complications and age of patients, tumor size, tumor volume, hypertension, or diabetes mellitus.

In addition to a significant relationship between radiation-induced complications and distance to the optic nerve disc or macula, there was a trend toward increase in radiation-induced complications (retinopathy or papillopathy) associated with maximal sclera dose of more than 500 Gy (at 2 mm from plaque surface) although this was not statistically significant (Table 2).

DISCUSSION

Radioactive plaque radiotherapy has become an important alternative for the treatment of ocular melanoma (1, 2). Several retrospective studies have reported local control of choroidal melanoma in 81–100% of cases at mean follow-up times from 34 months to 64 months (3–19). The differences in the results from these retrospective series are due to variations in the tumor size, location, plaques, dose, and prescription methods. The COMS Group performed a prospective, randomized trial to compare efficacy of episcleral plaque radiotherapy with conventional nucleation for medium-size ocular melanomas. Uniform plaques and identical doses and methods of prescription were adopted in this multi-institutional trial. The preliminary result of this study showed the 5-year survival rates to be 81% and 82% for surgery and brachytherapy, respectively (20, 21). Our ret-

rospective series has shown a progression-free survival of 93% at 5 years, with a median follow-up of 49 months.

There are major differences between the COMS plaques and the Nag plaques. The COMS study uses five circular plaques (of 12, 14, 16, 18, and 20 mm diameter, respectively), whereas the Nag plaques come in various shapes and sizes to fit individual tumors. Initially, the plaques were custom-made for each tumor. However, we now have a large selection of plaques to fit most tumors. Occasionally, we still have to custom make a plaque to fit an irregular shaped tumor. The COMS guidelines require the use of a plaque to cover the tumor with a minimum of 2 mm on all sides to account for microscopic extension of melanoma tumors into the surrounding choroid and for errors in placement. We use only a 1-mm margin because the base of the tumor always gets a much higher dose than the prescribed dose and therefore is not underdosed. Further, the use of unrimmed plaques extends the dose distribution beyond the physical edge of the plaque, hence a smaller margin can be used.

Retinal damage is a function of radiation dose and more importantly the area of retina irradiated. We therefore consider that it is very important to reduce the area of retina that is irradiated. Figure 7 illustrates the differences between a COMS plaque and Nag plaque for the irradiation of a noncircular (15 × 5 mm) tumor. A 17 × 7 mm oval Nag plaque will be required, whereas a 20-mm-diameter, circular plaque will be required, whereas a 20-mm-diameter, circular

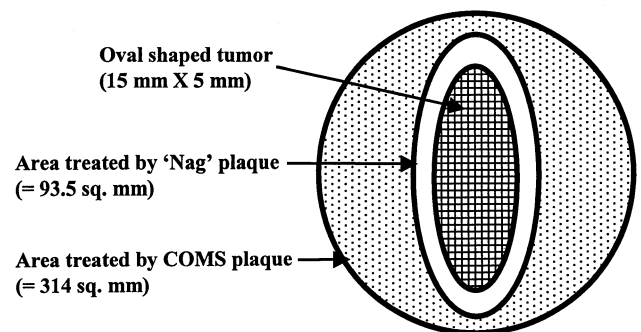


Fig. 7. Comparison between Collaborative Ocular Melanoma Study (COMS) plaque and Nag plaque for the irradiation of a noncircular (15 × 5 mm) tumor. A 17 × 7 mm oval Nag plaque will be required, whereas a 20-mm-diameter, circular plaque would be used if treated by a COMS plaque. Note that the Nag plaque irradiates a far smaller area of the normal sclera/retina (93.5 mm²) versus the COMS plaque (314 mm²).

Table 2. Relationship between maximum scleral dose and radiation-induced complications

Max sclera dose (Gy)	Number of patients	Number of retinopathy or papillopathy	% retinopathy or papillopathy
<500	44	15	34
>500	34	16	47
>600	26	13	50
>700	13	6	46
>800	7	4	57
>1000	4	3	75
Total	78	31	40

lar plaque would be used if treated by a COMS plaque. The Nag plaque would irradiate 93.5 mm² of normal sclera/retina, whereas the COMS plaque would irradiate 314 mm² of normal retina (i.e., 240% greater). Although the incidence of scleral necrosis can be expected to be related to the dose to the sclera, the amount of visual loss is dependent on the area of normal retina irradiated. Hence, visual loss is expected to be lower with the use of the Nag plaque. There was some initial fear and reservation that there might be a higher tumor recurrence with the use of our smaller plaques. However, this study has shown that there is a very low recurrence rate associated with our smaller plaques. An additional feature of the Nag plaque is that the inner surface of our plaque is raised 2 mm from the surface of sclera to reduce the radiation dose to sclera and can potentially lead to reduction in radiation-related complications.

The COMS protocol prescribes the dose on the central axis at the tumor apex if the tumor height is more than 5 mm. If the tumor is less than 5 mm in height, the prescription point is to a point on the central axis 5 mm from the inner surface of the sclera. The dose prescription method used in our study is different from that used in other institutions, including the COMS protocol. The prescription point in our study (Nag's prescription point) is located at the same height as the apex of the tumor but laterally, perpendicular to the edge of the choroidal melanoma in its greatest basal dimension (rather than prescribing to the apex of the tumor at the central axis) (Fig. 5). Doing so prevents any underdosage, especially for a flat or collar stud-type of tumor in which the apex of the tumor is a plateau, not dome shaped. Our prescription method ensures that the prescribed dose is delivered to the entire tumor irrespective of the shape of the tumor; however, it delivers a dose that is higher than the prescription dose at the apex of the tumor at central axis. Because of our unique prescription method, the tumors in our series received doses ranging from 33 to 147 Gy (median 85 Gy) if the doses were measured by the COMS method (i.e., at 5 mm for tumor height <5 mm and at the apex for tumors >5 mm in height). For small diameter tumors that are less than 5 mm in height, our prescription method results in delivery of a lower dose than would have been given if the COMS prescription method were used. For example, for a tumor of 1.9 mm height and 7 mm diameter, 85 Gy by our prescription method was equivalent to giving 47.6 Gy by the COMS method (i.e., at 5 mm height along central axis). However, we had excellent control of these smaller tumors at these lower doses. We believe that the results of our present study (low incidence of local progression and relatively low incidence of retinopathy or papillopathy) justify our method of dose prescription. The situation was reversed for large diameter tumors greater than 5 mm in height. For example, for a tumor of 7.4 mm in height and 12 mm diameter, 85 Gy by our prescription method was equivalent to giving 115.7 Gy by the COMS method (i.e., at the apex along the central axis). Because the COMS study has shown control of tumor with 85 Gy prescribed to the apex along the central axis, it is possible that we may be able

to lower our doses for the larger tumors, even though we did not have statistically increased retinopathy or papillopathy with the higher apical doses used in this study. In this regard, it is to be noted that the guidelines recently issued by the American Brachytherapy Society (ABS) recommend an ¹²⁵I prescription dose of 85 Gy (at 0.60–1.05 Gy/h) to pass through the apex and encompass the entire tumor (23). However, the ABS also noted that, because of the steep dose gradient that exists in ocular brachytherapy, it may be able to control the larger ("taller") tumors with lower doses if the dose is prescribed to the apex and that further research is warranted to establish the optimum dose and prescription method for the treatment of uveal melanomas (23).

Potter *et al.* (19) recently reported their experience with eye plaque irradiation for 93 juxtapapillary choroidal melanoma. They reported 15% local failure at mean time of 41 months and a 12% rate of distant failure after a mean interval of 44 months. The predictive factors for their local failures were the age of the patients (<35 years) and the superior or inferior location of the tumor. Predictive factors for distant failure were large tumor size (basal diameter >6.0 mm), the superior location of the tumor, and local tumor recurrence. However, in our study, only four local failures had occurred after a mean time of 47 months, and one distant failure occurred only 1 month after radiotherapy. Reliable predictive factors cannot be derived from our data because of this low incidence of failure.

In the present study of 78 patients, 31 patients (40%) developed radiation-associated complications. However, in the subgroup of 22 patients in whom the distance from tumor to macula or optic nerve disc was <2 mm, 54% developed moderate to severe retinopathy or papillopathy (statistically significant on a univariate level). In addition, only 32% of this subgroup of 22 patients maintained their useful vision (20/200 or better). Hence, patients with tumors close to the macula or optic disc should be informed of the poor visual prognosis before the treatment. There were no statistical correlations between radiation-associated complications and the age of the patient (>45 years), medical history of diabetes mellitus, or tumor size, as opposed to prior observations (1, 2, 19). The lack of statistical difference could be due to the small number of patients younger than 45 years old ($n = 7$), the small number of patients with diabetes mellitus ($n = 9$), or the small size of this study (78 patients).

There are some limitations to our study. As with all retrospective, nonrandomized studies, these results should be interpreted cautiously. Although a median follow-up time of 49 months (mean time 44 months) is reasonably appropriate, more recurrences and further late visual complications may emerge with continued follow-up. Although statistical analysis requires a large number of cases for meaningful subset analysis of predictive factors for recurrence or complications, there were only 78 patients in this study. There is also the inherent bias that this study took place during the course of the ongoing COMS study. The patients reported here either had refused enucleation, refused or were ineligible to participate in the

COMS study, an unfavorable tumor location (such as juxta-papillary tumor, tumor with optic nerve infiltration), or a previous history of malignancy or poor medical condition unsuitable for enucleation. Hence, they constituted a worse prognostic group. Finally, our analysis did not consider histopathologic variables, which may affect outcome. Despite these limitations, the results of our study are impressive, with acceptable morbidity and a low incidence of local and distant failure.

CONCLUSION

Our experience suggests that custom-made Nag plaque radiotherapy can control medium-size choroidal melanoma. A 1-mm margin (rather than the standard 2-mm minimum tumor margin) may be sufficient when this plaque and prescription methods are used. Patients with tumors close (≤ 2 mm) to optic disc or macula have a significantly poorer visual outcome.

REFERENCES

1. Finger PT. Radiation therapy for choroidal melanoma (therapeutic review). *Surv Ophthalmol* 1997;42:215–232.
2. Fontanesi J, Nag S. Brachytherapy for ocular disease. In: Nag S, editor. Principles and practice of brachytherapy. Armonk, NY: Futura Publishing Company Inc.; 1997. p. 291–304.
3. Finger PT, Buffa A, Mishra S, *et al.* Palladium-103 plaque radiation therapy for uveal melanoma. Clinical experience. *Ophthalmology* 1994;101:256–263.
4. Finger PT, Moshfeghi DM, Ho TK. Palladium-103 ophthalmic plaque radiation therapy. *Arch Ophthalmol* 1991;109:1610–1613.
5. Lommatzsch PK. Treatment of choroidal melanoma with $^{106}\text{Ru}/^{106}\text{Rh}$ beta-ray applicators. *Surv Ophthalmol* 1974;19:85–100.
6. Lommatzsch PK. $^{106}\text{Ru}/^{106}\text{Rh}$ plaque radiotherapy for malignant melanomas of the choroid. With follow-up results more than 5 years. *Doc Ophthalmologica* 1988;68:225–238.
7. Fontanesi J, Meyer D, Shizhao X, *et al.* Treatment of choroidal melanoma with I-125 plaque. *Int J Radiat Oncol Biol Phys* 1993;26:619–623.
8. Sealy R, Buret E, Cleminshaw H, *et al.* Progress in the use of iodine therapy for tumors of the eye. *Br J Radiol* 1980;53:1052–1060.
9. Stallard HB. Radiobiology for malignant melanoma of the choroid. *Br J Ophthalmol* 1966;50:147–155.
10. Davidorf FH, Makley TA, Lang JR. Radiotherapy of malignant melanoma of the choroid. *Trans Am Acad Ophthalmol Otolaryngol* 1976;81:849–861.
11. Davidorf FH, Paika JT, Makley TA, *et al.* Radiotherapy of choroidal melanoma: An 18-year experience with radon. *Arch Ophthalmol* 1978;105:352–355.
12. Garretson BR, Robertson DM, Earle JD. Choroidal melanoma treatment with ^{125}I brachytherapy. *Arch Ophthalmol* 1987;105:1394–1397.
13. Heikkonen J, Summanen P, Immonen L, *et al.* Radiotherapy of malignant melanoma of the uvea with ^{125}I seeds. *Acta Ophthalmol* 1992;70:780–785.
14. Lean EK, Cohen DM, Liggett PE, *et al.* Episcleral radioactive plaque therapy: Initial clinical experience with 56 patients. *Am J Clin Oncol* 1990;13:185–190.
15. Lommatzsch PK. Results after beta-irradiation ($^{106}\text{Ru}/^{106}\text{Rh}$) of choroidal melanomas: 20 years experience. *Br J Ophthalmol* 1986;70:844–885.
16. Packer S. Iodine-125 radiation of posterior uveal melanoma. *Ophthalmology* 1987;94:1621–1626.
17. Packer S, Rotman M, Salanito P. Iodine-125 irradiation of choroidal melanoma. Clinical experience. *Ophthalmology* 1984;91:1700–1708.
18. Packer S, Stoller S, Lessor ML, *et al.* Long-term results of ^{125}I irradiation of uveal melanoma. *Ophthalmology* 1992;99:767–774.
19. Potter PD, Shields CL, Shields JA, *et al.* Plaque radiotherapy for juxta-papillary choroidal melanoma. *Arch Ophthalmol* 1996;114:1357–1365.
20. The Collaborative Ocular Melanoma Study Group. The COMS randomized trial of Iodine-125 brachytherapy for choroidal melanoma, II: Characteristics of patients enrolled and not enrolled. *Arch Ophthalmol* 2001;951–961.
21. The Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine-125 brachytherapy for choroidal melanoma, III: Initial mortality findings. *Arch Ophthalmol* 2001;969–982.
22. Nath R, Anderson LL, Luxton G, *et al.* Dosimetry of interstitial brachytherapy sources: Recommendations of the AAPM Radiation Committee Task Group No. 43. *Med Phys* 1995;22:209–234.
23. Nag S, Quivey JM, Earle JD, *et al.* The American Brachytherapy Society recommendations for brachytherapy of uveal melanomas. *Int J Radiat Oncol Biol Phys* 2003;56:544–555.